

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

22nd November 2024

**SP7 - General Insurance Reserving and Capital
Modelling**

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

- Q. 1)** You are an actuary of an insurance company that writes large industrial property business with large deductibles. It is subject to large losses, which are usually reported quickly. Reserves are always set up in respect of these losses, but these are often found to be excessive or inadequate. You are concerned about estimation of ultimate claims from this line of business, which has been carried out using chain ladder method and often has been unstable in the past. It has been suggested that the use of Bornhuetter- Ferguson methods would provide an element of stability in reserve estimates.

Discuss the suggestion.

[5 Marks]

- Q. 2)** A large insurance company writing multiple line of business recently decide to stop writing a particular retail line of business due to huge losses year on year. The company does not foresee a profitable market and foresees a lot of uncertainty in this line of business at least for next few years.

The line of business contributes approximately 15% of total written premium for the last financial year and has a long tail.

The company is considering the below 2 options:

Option 1: run-off the existing liabilities

Option 2: sell the run-off portfolio to another insurer/reinsurer

- a)** Discuss the possible options, its advantages & disadvantages for the company.

(11)

- b)** Suggest any other method of transferring reserve development risks in addition to mentioned above. Also mention its features.

(5)

[16 Marks]

- Q. 3)** You have been given the following summary information, extracted from the accounts of a general insurance company. All Figures are in INR Crores.

Profit & Loss Account for the year ended 31 March 2024

Balance Sheet as at 31 March 2024

Gross Written Premium	1720
Less : Reinsurance ceded	825
Net Written Premium	895

Gross Earned Premium	1500
Less : Reinsurance ceded	720
Net Earned Premium	780

Gross Claims Incurred	2160
Less : Reinsurance recovery	1350
Net Claims Incurred	810

Gross Commission Paid	340
Less : Commission Earned	260
Net Commission Paid	80

Investment Income	630
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Other Expenses of Management	180
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Gross Profit	340
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Technical Provisions Net of

Reinsurance	
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UPR	330
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Outstanding Claims	1425
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IBNR	4545
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Total	6300
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Paid up Share Capital	200
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Retained Profits	1300
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	<u>7800</u>
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Investments

Equity	700
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Fixed Interest Securities	
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<4 year duration	2500
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>4 year duration	4000
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Land & Buildings	200
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Cash & Bank Balances	50
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Other Current Assets	350
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Less : Taxation	100	
Net Profit	240	
		7800

Discuss the nature of the business underwritten by company and its financial condition, as it can be determined from these accounts. **[10 Marks]**

Q. 4) A junior analyst in your team has just returned from a meeting with the claims manager. The claims manager anticipates the average term of loss to settlement of claims to reduce from 4 years to 2 years. The claims manager explains this is due to a very aggressive and proactive approach that they are taking to claims. Assume that you adopted a ACPC (Average Claim per count) method for this class of business in your previous valuation exercise.

- a) Determine the potential impact on the overall valuation results at the next valuation exercise. (6)
- b) Distinguish the further information you would suggest the junior analyst obtain. (2)
- c) Explain in few words adjusting the valuation method to incorporate the change. (2)
- d) Propose potential reasons, other than aggressive/proactive claims management style, why the average term of a claims might reduce from 4 years to 2 years for a particular class of insurance business. (6)

[16 Marks]

Q. 5) The regulator in a particular country has decided to move from a regulatory deterministic model to stochastic model to determine its regulatory capital requirement.

- a) Suggest possible reasons why the regulator may be considering a stochastic model rather than a deterministic model to determine its capital. (6)

A general insurer in this particular country has written personal lines building insurance covering home and its content. It currently reinsures business on a loss – occurring basis.

- b) Describe how the insurer will determine the insurance risk for this class of business. You must consider the different claim types. (12)

- c) List the practical issues the general insurer must consider while building a capital model. (6)

[24 Marks]

Q. 6) A Company established 20 years ago is setting its reserves. It uses a statistical modelling software to set up its reserves. In the past years while setting up the reserves, it was observed that net cashflows outgo do not match with those expected and exceeds as compared to current reserves due to various uncertainties.

State seven risks & uncertainties with its definition and at least one example of each that the company may be facing at this point of time.

Suggest one way in which this uncertainty can be removed and a more certain reserve levels can be achieved. **[8 Marks]**

Q. 7) ABC Insurance company has been writing third party liability insurance since 2017. The company has not changed its underwriting standards for this portfolio for many years.

Consider the paid claims data given below for past 7 years for this line of business.

AY/DY	12	24	36	48	60	72	84	96	EP
2017	200	17,500	38,500	44,500	48,800	52,700	54,900	54,900	96,200
2018	600	18,300	41,100	49,500	54,000	56,900	57,300		93,600
2019	600	18,700	50,900	60,900	64,500	66,700			1,05,400
2020	300	15,300	52,700	64,100	69,700				1,20,100
2021	400	15,200	55,800	65,200					1,36,000
2022	400	14,800	50,700						1,41,000
2023	400	16,500							1,44,800
2024	200								1,68,900

- a) Apply Chain ladder method to arrive at the Ultimate Claims and thus Ultimate Claims Ratio for all the Accident years. Mention the assumption used. (7)
- b) Explain the issues with the method above. (2)
- c) The Senior colleague asked you to apply Bornhuetter-Ferguson Method to arrive at the Ultimate Claims & Ultimate Claims Ratio. The Senior colleague also asked you to use underwriter's judgement of Expected Loss Ratio of 60%. (4)
- d) State one disadvantage of Bornhuetter-Ferguson Method based on question in sub part c). (1)
- e) State one another alternative method that can be used in place of Bornhuetter-Ferguson Method which takes care of above disadvantage. (1)

[15 Marks]

- Q. 8)** In country ABC, the current practice of accounting for multi-year policies was to account for total GWP in the year in which policy has been underwritten. The regulator has recently come up with a regulation which asks the companies to divide the GWP in case of multi-year policies into number of years (1/n) for which the policy has been written and the GWP be accounted only for 1 year in current year.

The regulation in the country does not allow to defer the acquisition costs. It also does not allow the companies to exceeds its total expenses as a proportion of underwritten premium beyond a prescribed limit. Few companies in the country have been given the option to account 50% of total written premium as earned premium in the same year and 50% in the next accounting year. All the other companies account earns the written premium on linear basis during the exposure period of the policy.

Describe the possible impact on the company XYZ operating in Country ABC. Company XYZ has opted for the option to earn 50% of written premium in the same accounting year and earn remaining 50% in the next accounting year.

[6 Marks]
